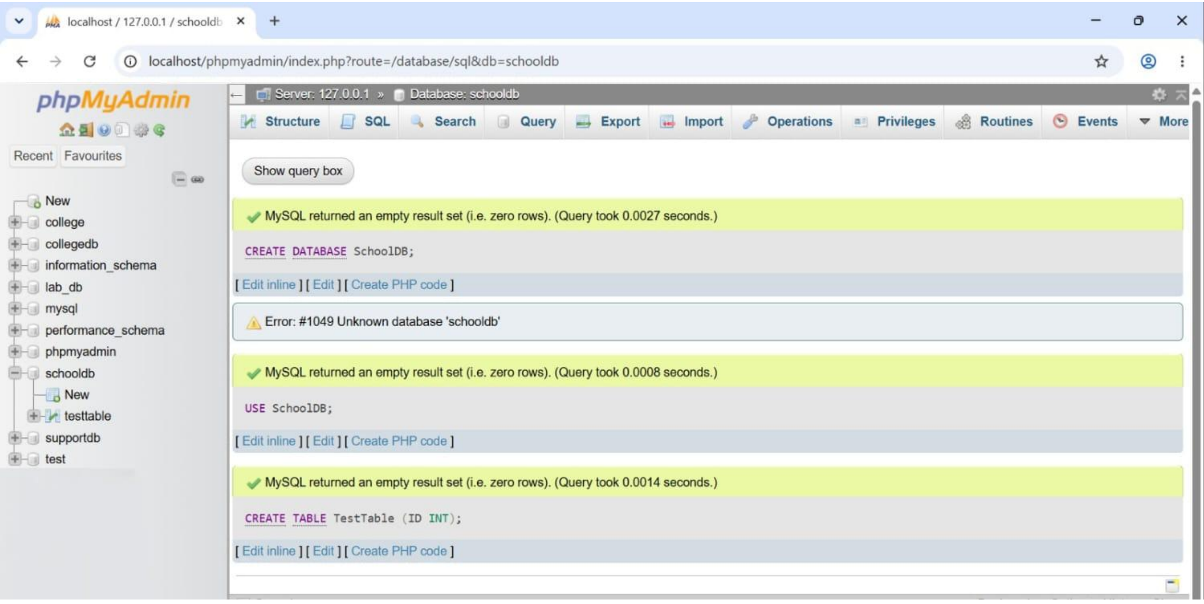
ASSIGNMENT NO. 1: Introduction to Databases and SQL Environment

Questions:

Q1) Write SQL commands to create a database named SchoolDB.
The program should:
1. Create a new database named SchoolDB.
2. Select the database for use.
3. Display confirmation by creating a simple table inside it.

Query-
CREATE DATABASE SchoolDB;
USE SchoolDB;
CREATE TABLE TestTable (ID INT);

Output-


Q2) Create a table named Student with the following columns:

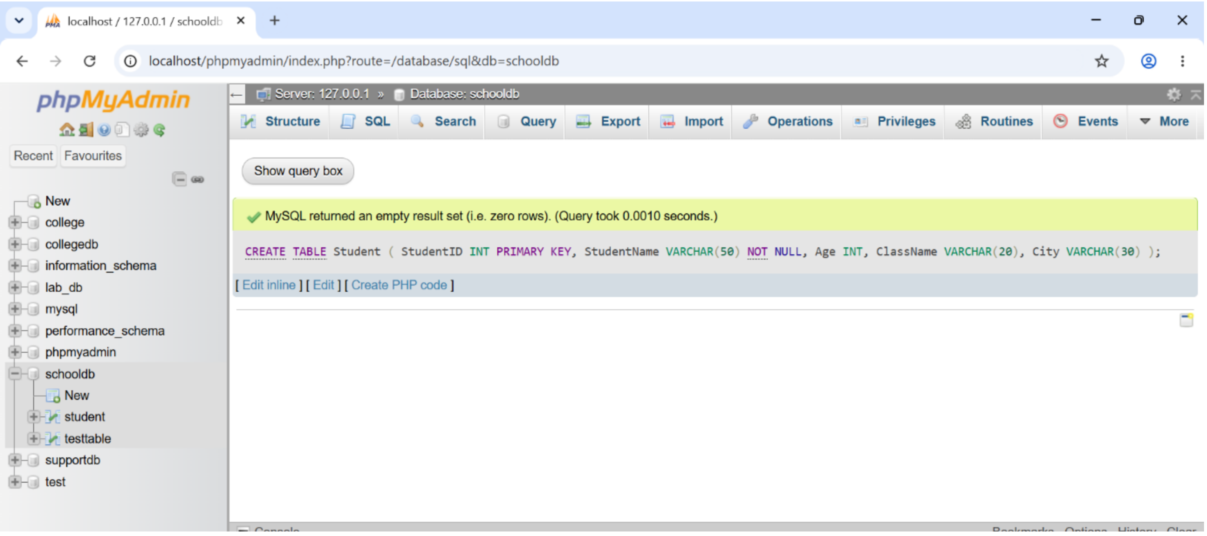
Column Name	**Data Type**	**Constraint**
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
Age	INT	
ClassName	VARCHAR(20)	
City	VARCHAR(30)	

The program should:
1. Create the table.
2. Define proper data types.
3. Apply primary key on StudentID.

Query-
CREATE TABLE Student (
StudentID INT PRIMARY KEY,

    StudentName VARCHAR(50) NOT NULL,
    Age INT,
    ClassName VARCHAR(20),
    City VARCHAR(30)
);

Output-



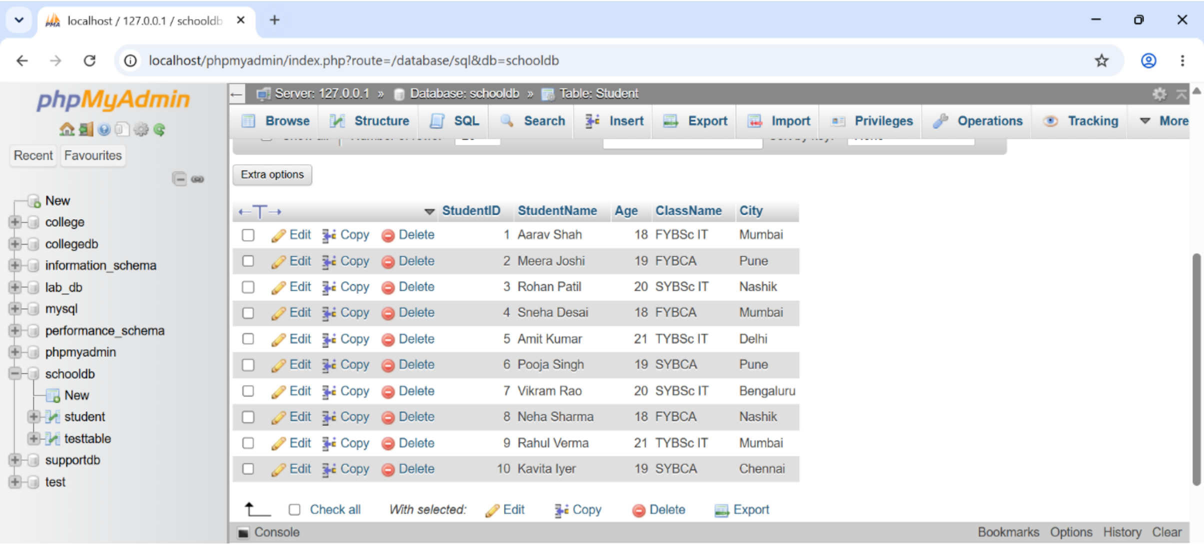
Q3) Insert at least 10 records into the Student table.
The records should include:
1. Student ID
2. Student name
3. Age
4. Class name
5. City
After inserting data, display all records using the SELECT command.

Query-

INSERT INTO Student (StudentID, StudentName, Age, ClassName, City)
VALUES
 (1, 'Aarav Shah', 18, 'FYBsc IT' , 'Mumbai'),
 (2, 'Meera Joshi', 19, 'FYBCA' , 'Pune'),
 (3, 'Rohan Patil', 20, 'SYBsc IT' , 'Nashik'),
 (4, 'Sneha Desai', 18, 'FYBCA' , 'Mumbai'),
 (5, 'Amit Kumar', 21, 'TYBsc IT' , 'Delhi'),
 (6, 'Pooja Singh', 19, 'SYBCA' , 'Pune'),
 (7, 'Vikram Rao', 20, 'SYBsc IT' , 'Bengaluru'),
 (8, 'Neha Sharma', 18, 'FYBCA' , 'Nashik'),
 (9, 'Rahul Verma', 21, 'TYBsc IT' , 'Mumbai'),
 (10, 'Kavita Iyer', 19, 'SYBCA' , 'Chennai');

 SELECT * FROM Student;

Output-



Q4) Create a table named Course with the following fields:

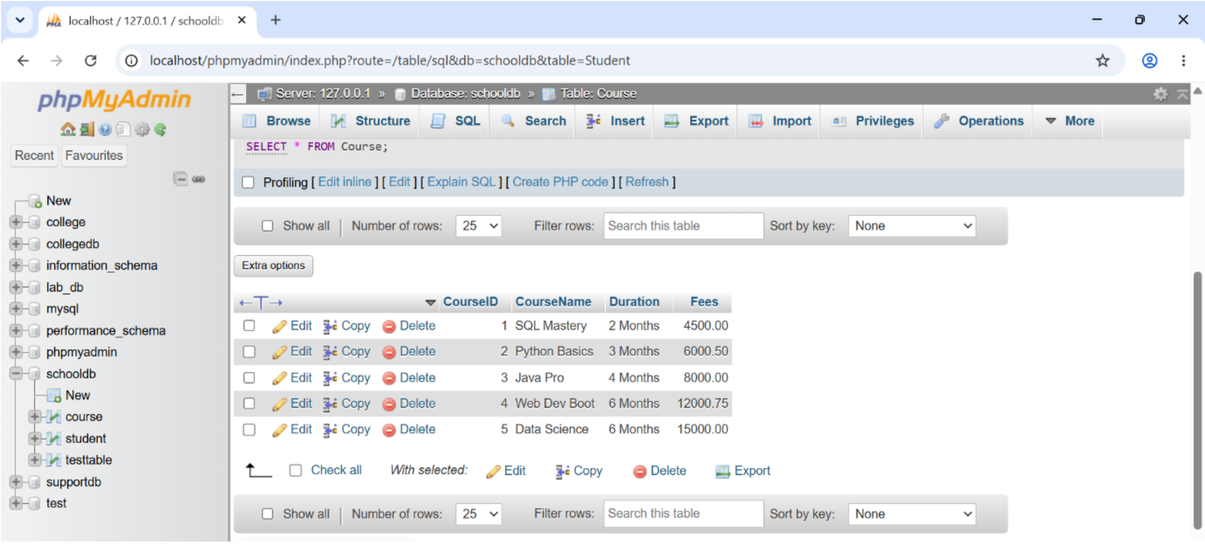
Column Name	**Data Type**
CourseID	INT
CourseName	VARCHAR(50)
Duration	VARCHAR(20)
Fees	DECIMAL(10,2)

The program should:
1. Create the table.
2. Add a primary key on CourseID.
3. Insert at least 5 course records.
4. Display all course records.

Query-

```
CREATE TABLE Course (
  CourseID INT PRIMARY KEY,
  CourseName VARCHAR(50),
  Duration VARCHAR(20),
  FEES DECIMAL(10,2)
);

INSERT INTO Course (CourseID, CourseName, Duration, Fees) VALUES
  (1 , 'SQL Mastery' , '2 Months' , 4500.00),
  (2 , 'Python Basics' , '3 Months' , 6000.50),
  (3 , 'Java Pro' , '4 Months' , 8000.00),
  (4 , 'Web Dev Boot' , '6 Months' , 12000.75),
  (5 , 'Data Science' , '6 Months' , 15000.00);

SELECT * FROM Course;
```

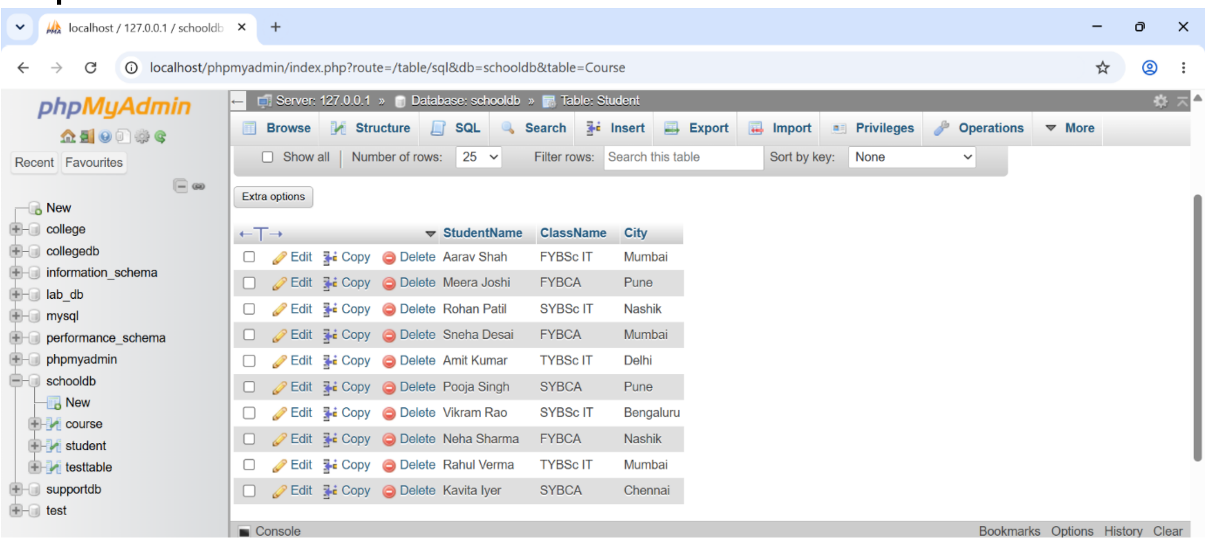
Output-



Q5) Write SQL queries to display only selected columns from the Student table.
Display:
1. StudentName
2. ClassName
3. City
Expected output should show only these three columns.

Query-
SELECT StudentName, ClassName, City
FROM Student;

Output-



Q6) Write SQL commands to view the structure of the Student table.
Use the suitable command according to the database environment:

Database	**Command**
MySQL	DESC Student;
PostgreSQL	tudent

SQLite	PRAGMA table_info(Student);

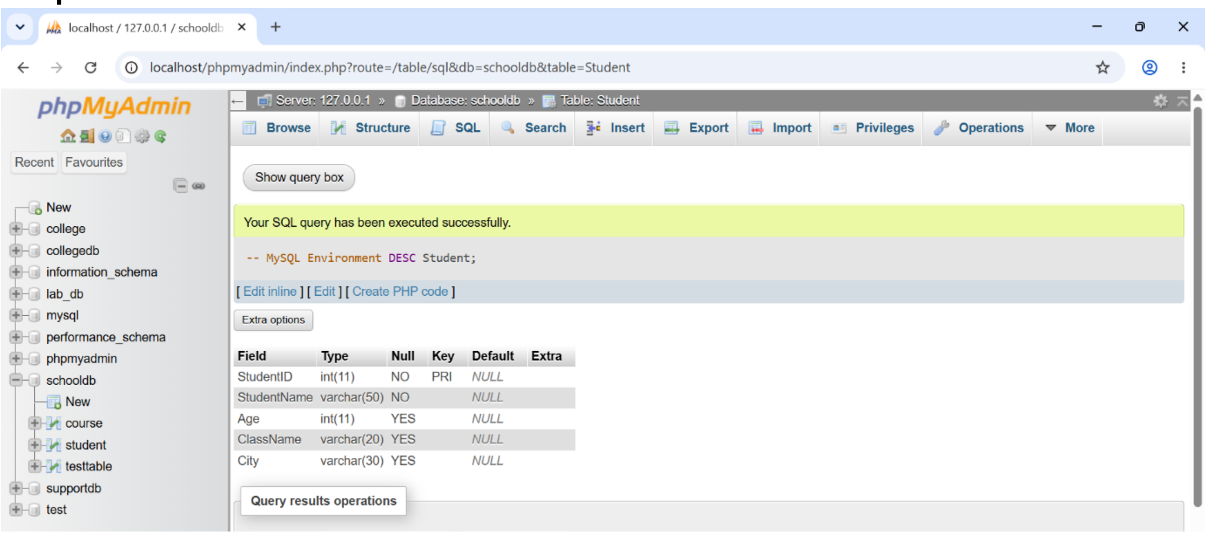
Query-

-- MySQL Environment
DESC Student;

-- PostgreSQL Environment
\d Student

-- SQLite Environment
PRAGMA table_info(Student);

Output-



Q7) Prepare a CSV file named students.csv and import it into the Student table.
The CSV file should contain:

StudentID	**StudentName**	**Age**	**ClassName**	**City**
1	Aarav Shah	18	FYBSc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBSc IT	Nashik

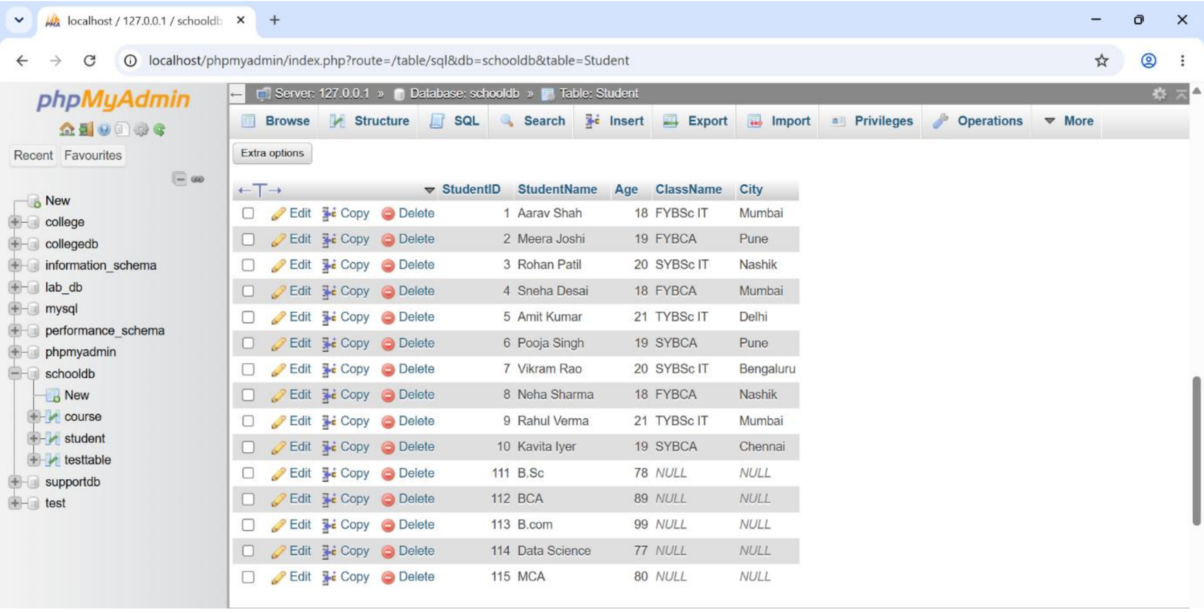
The task should include:
1. Creating a matching table.
2. Importing the CSV file.
3. Displaying imported records using SELECT.

Query-

LOAD DATA INFILE ‘Students.csv’
INTO TABLE Student
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;

SELECT * FROM Student;

Output-



Q8) Create a database named EmployeeDB and a table named Employee.
The table should contain:

Column Name	Data Type
EmployeeID	INT
EmployeeName	VARCHAR(50)
Department	VARCHAR(30)
Salary	DECIMAL(10,2)
City	VARCHAR(30)

The program should:
1. Create the database.
2. Create the Employee table.
3. Insert 8 employee records.
4. Display all employee records.

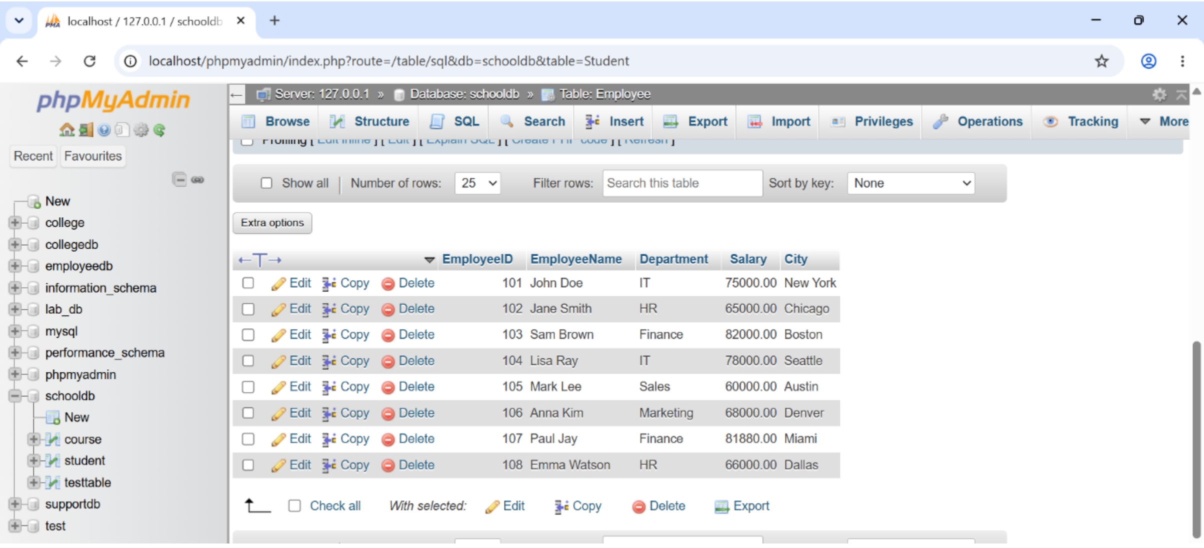
Query-

```
CREATE DATABASE EmployeeDB;
USE EmployeeDB;

CREATE TABLE Employee (
  EmployeeID INT PRIMARY KEY, EmployeeName VARCHAR(50),
  Department VARCHAR(30), Salary DECIMAL(18,2), City VARCHAR(30)
);

INSERT INTO Employee VALUES
  (101, 'John Doe', 'IT', 75000.00, 'New York'),
  (102, 'Jane Smith', 'HR', 65000.00, 'Chicago'),
  (103, 'Sam Brown', 'Finance', 82000.00, 'Boston'),
```

(104, 'Lisa Ray', 'IT', 78000.00, 'Seattle'),
(105, 'Mark Lee', 'Sales', 60000.00, 'Austin'),
(106, 'Anna Kim', 'Marketing', 68000.00, 'Denver'),
(107, 'Paul Jay', 'Finance', 81880.00, 'Miami'),
(108, 'Emma Watson', 'HR', 66000.00, 'Dallas');

SELECT * FROM Employee;

Output-

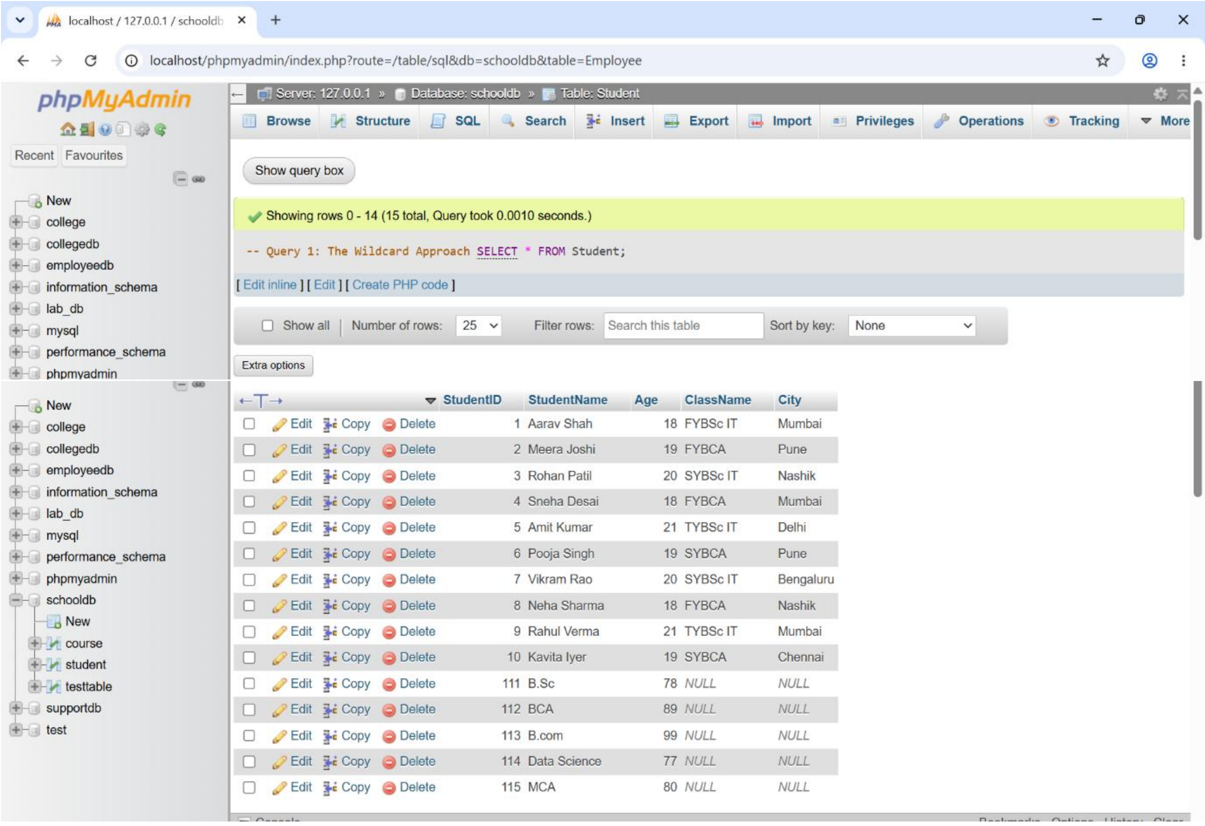


EmployeeID	EmployeeName	Department	Salary	City
101	John Doe	IT	75000.00	New York
102	Jane Smith	HR	65000.00	Chicago
103	Sam Brown	Finance	82000.00	Boston
104	Lisa Ray	IT	78000.00	Seattle
105	Mark Lee	Sales	60000.00	Austin
106	Anna Kim	Marketing	68000.00	Denver
107	Paul Jay	Finance	81880.00	Miami
108	Emma Watson	HR	66000.00	Dallas

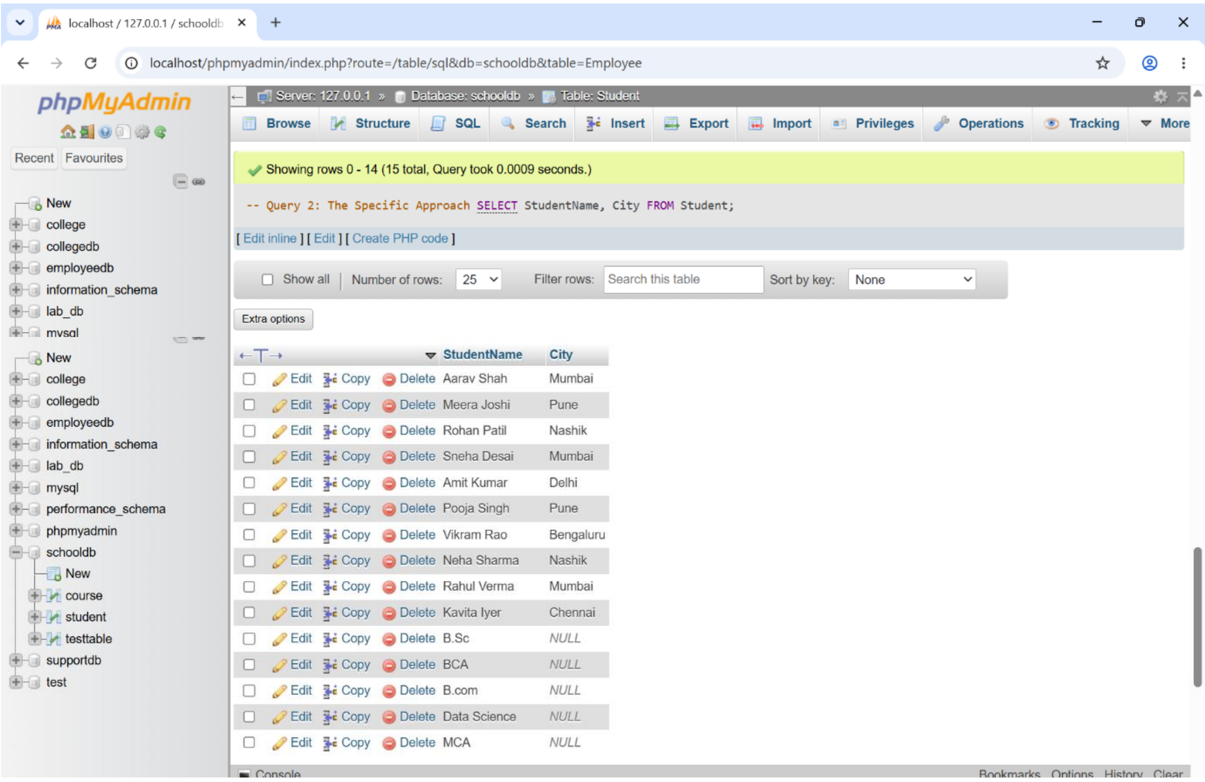
Q9) Write SQL queries to compare the difference between SELECT * and selecting specific columns.
The task should include:
1. Query using SELECT * FROM Student;
2. Query using SELECT StudentName, City FROM Student;
3. Write two lines explaining the difference between both outputs.

Query-
--query 1: the wildcard approach
Select *
FROM Student
--query2: the specific approach
Select StudentName, City
From student;

Output-



StudentID	StudentName	Age	ClassName	City
1	Aarav Shah	18	FYBSc IT	Mumbai
2	Meera Joshi	19	FYBCA	Pune
3	Rohan Patil	20	SYBSc IT	Nashik
4	Sneha Desai	18	FYBCA	Mumbai
5	Amit Kumar	21	TYBSc IT	Delhi
6	Pooja Singh	19	SYBCA	Pune
7	Vikram Rao	20	SYBSc IT	Bengaluru
8	Neha Sharma	18	FYBCA	Nashik
9	Rahul Verma	21	TYBSc IT	Mumbai
10	Kavita Iyer	19	SYBCA	Chennai
111	B.Sc	78	NULL	NULL
112	BCA	89	NULL	NULL
113	B.com	99	NULL	NULL
114	Data Science	77	NULL	NULL
115	MCA	80	NULL	NULL



StudentName	City
Aarav Shah	Mumbai
Meera Joshi	Pune
Rohan Patil	Nashik
Sneha Desai	Mumbai
Amit Kumar	Delhi
Pooja Singh	Pune
Vikram Rao	Bengaluru
Neha Sharma	Nashik
Rahul Verma	Mumbai
Kavita Iyer	Chennai
B.Sc	NULL
BCA	NULL
B.com	NULL
Data Science	NULL
MCA	NULL

Q10) Create a complete SQL script for a mini college database.
The script should include:
1. Creating database MiniCollegeDB.

2. Creating tables Student, Course, and Faculty.
3. Adding suitable columns and data types.
4. Inserting at least 5 records in each table.
5. Displaying all records from each table.
6. Viewing the structure of each table.

Query-

CREATE DATABASE MiniCollegeDB;
USE MiniCollegeDB;

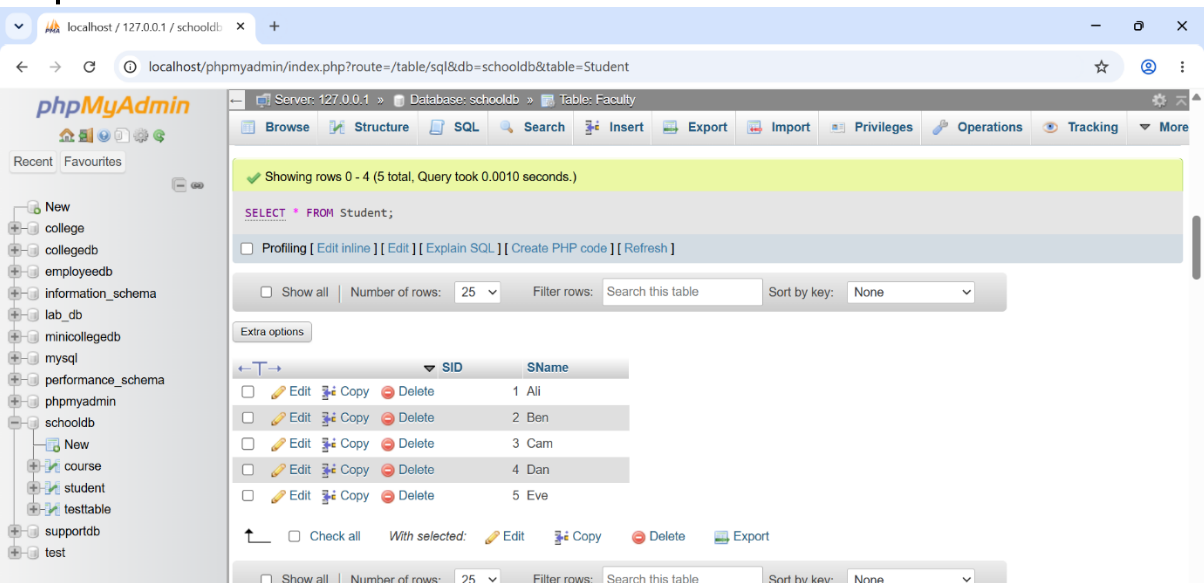
CREATE TABLE Student (SID INT PRIMARY KEY, SName VARCHAR(50));
CREATE TABLE Course (CID INT PRIMARY KEY, CName VARCHAR(50));
CREATE TABLE Faculty (FID INT PRIMARY KEY, FName VARCHAR(50));

INSERT INTO Student
VALUES (1,'Ali'), (2, 'Ben'), (3, 'Cam'), (4, 'Dan'), (5, 'Eve');
INSERT INTO Course
VALUES (10, 'Math'), (11, 'Sci'), (12, 'Art'), (13, 'Lit'), (14, 'CS');
INSERT INTO Faculty
VALUES (100, 'Dr.X'),(101, 'Dr.Y'),(102,'Dr.Z'), (103, 'Dr.A'), (104, 'Dr.B');

SELECT * FROM Student; SELECT * FROM Course; SELECT * FROM Faculty;

DESC Student; DESC Course; DESC Faculty;

Output



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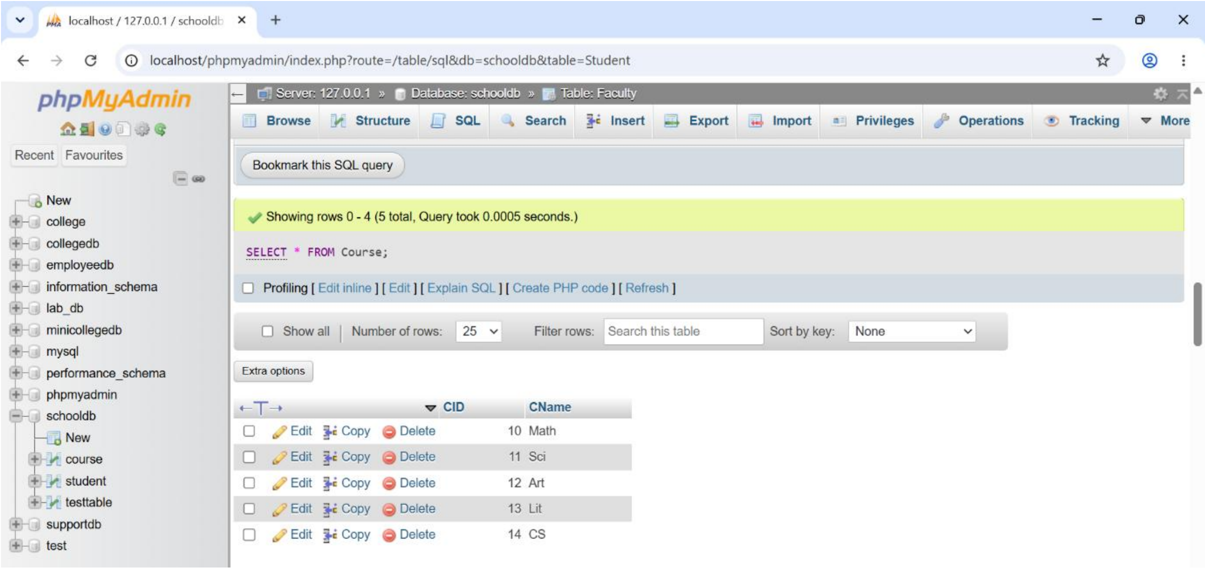
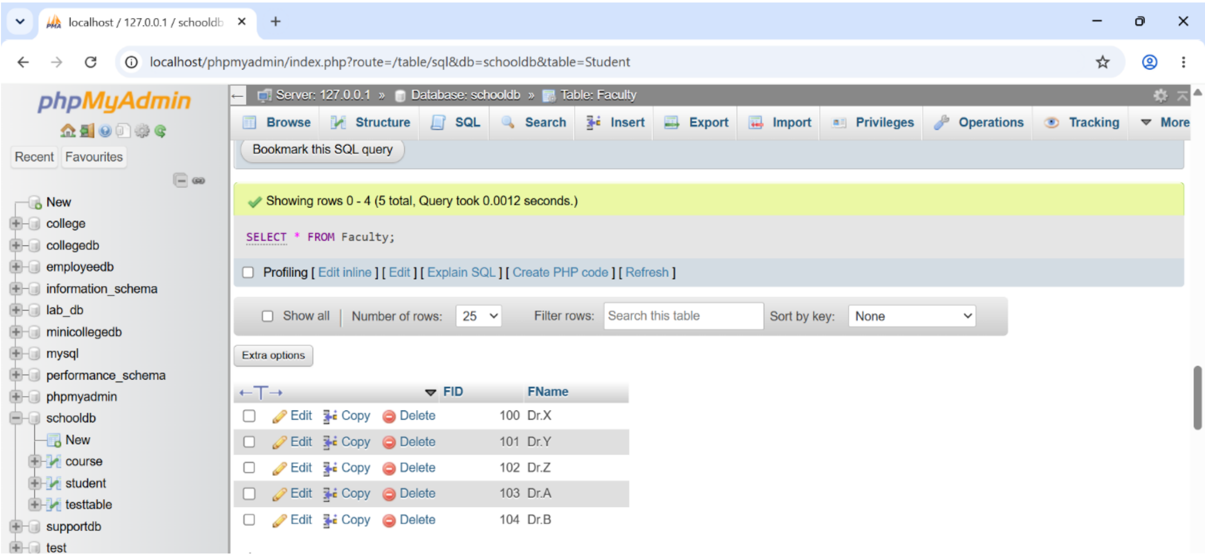
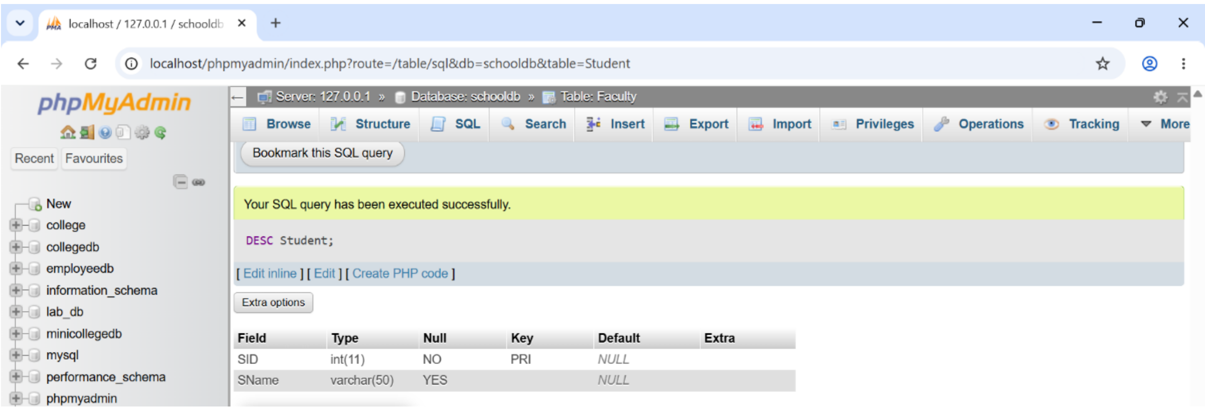

localhost / 127.0.0.1 / schooldb
localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student
phpMyAdmin
Server: 127.0.0.1 » Database: schooldb » Table: Faculty
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Bookmark this SQL query
Showing rows 0 - 4 (5 total, Query took 0.0005 seconds.)
SELECT * FROM Course;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
CID CName
Edit Copy Delete 10 Math
Edit Copy Delete 11 Sci
Edit Copy Delete 12 Art
Edit Copy Delete 13 Lit
Edit Copy Delete 14 CS
Recent Favourites
New
college
collegedb
employeedb
information_schema
lab_db
minicollegedb
mysql
performance_schema
phpmyadmin
schooldb
New
course
student
testtable
supportdb
test


localhost / 127.0.0.1 / schooldb
localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student
phpMyAdmin
Server: 127.0.0.1 » Database: schooldb » Table: Faculty
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Bookmark this SQL query
Showing rows 0 - 4 (5 total, Query took 0.0012 seconds.)
SELECT * FROM Faculty;
Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]
Show all | Number of rows: 25 Filter rows: Search this table Sort by key: None
Extra options
FID FName
Edit Copy Delete 100 Dr.X
Edit Copy Delete 101 Dr.Y
Edit Copy Delete 102 Dr.Z
Edit Copy Delete 103 Dr.A
Edit Copy Delete 104 Dr.B
Recent Favourites
New
college
collegedb
employeedb
information_schema
lab_db
minicollegedb
mysql
performance_schema
phpmyadmin
schooldb
New
course
student
testtable
supportdb
test


localhost / 127.0.0.1 / schooldb
localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student
phpMyAdmin
Server: 127.0.0.1 » Database: schooldb » Table: Faculty
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Bookmark this SQL query
Your SQL query has been executed successfully.
DESC Student;
[Edit inline] [Edit] [Create PHP code]
Extra options
Field Type Null Key Default Extra
SID int(11) NO PRI NULL
SName varchar(50) YES NULL
Recent Favourites
New
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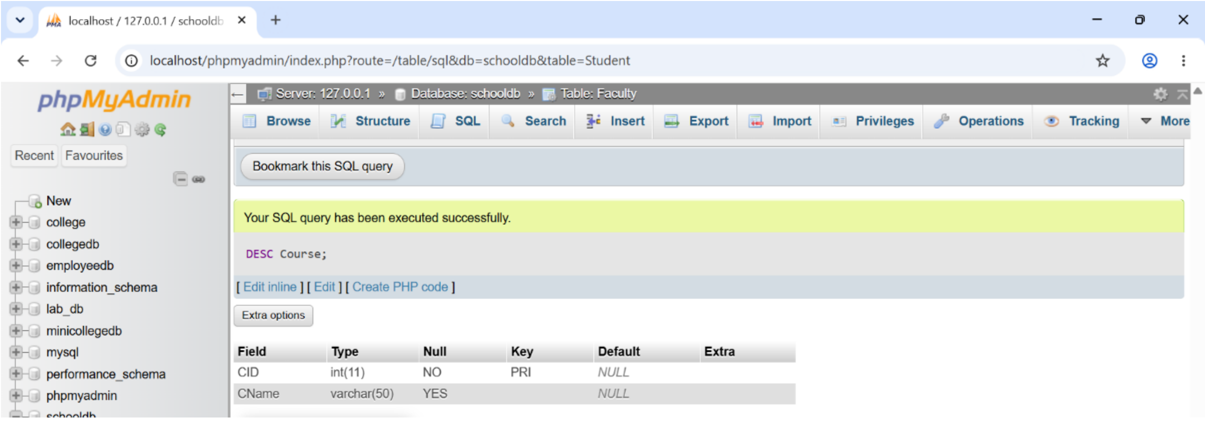
employeedb
information_schema
lab_db
minicollegedb
mysql
performance_schema
phpmyadmin


localhost / 127.0.0.1 / schooldb
localhost/phpmyadmin/index.php?route=/table/sql&db=schooldb&table=Student
phpMyAdmin
Recent Favourites
New
college
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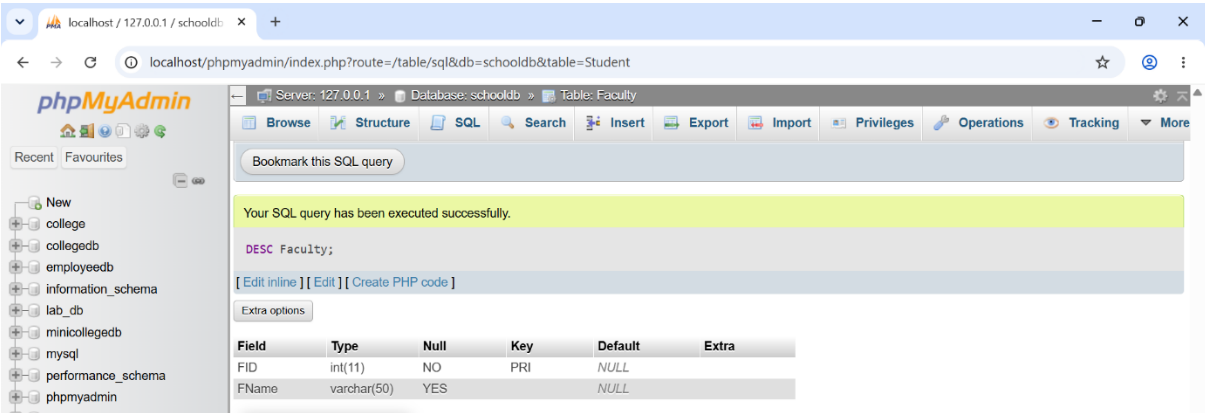
information_schema
lab_db
minicollegedb
mysql
performance_schema
phpmyadmin
Server: 127.0.0.1 » Database: schooldb » Table: Faculty
Browse Structure SQL Search Insert Export Import Privileges Operations Tracking More
Bookmark this SQL query
Your SQL query has been executed successfully.
DESC Course;
[Edit inline] [Edit] [Create PHP code]
Extra options
Field Type Null Key Default Extra
CID int(11) NO PRI NULL
CName varchar(50) YES NULL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Bookmark this SQL query
Your SQL query has been executed successfully.
DESC Faculty;
[Edit inline] [Edit] [Create PHP code]
Extra options
Field Type Null Key Default Extra
FID int(11) NO PRI NULL
FName varchar(50) YES NULL